2.6.4 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept below.

I need help. I got it. I could help someone.

I can write the converse of a mathematical statement.	
I can write the inverse of a mathematical statement.	
I can write the contrapositive of a mathematical statement.	
I can determine if a statement is biconditional.	
I can write and interpret an "all" statement.	
I can write and interpret a negation.	

Unit 2, Lesson 7: Proofs – Part 1

Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.

Learning Targets

- ☐ I can relate deductive reasoning to completing proofs based on assumptions and given statements.
- ☐ I can recognize paragraph, flow-chart, and two column proofs.

2.7.1 Warm-Up: Miami Art Deco

A style known as Art Deco began in Paris in the mid-1920s. Art Deco buildings are characterized by their geometric shapes and symmetry. Two of the most famous examples of Art Deco architecture are the Empire State Building and Chrysler Building in New York City.

Photos of three locations in Miami that use this style of architectural design are shown.

1. Draw a line of symmetry on each picture.

2. Which depiction of Art Deco do you find most interesting?

2.7.2 Exploration: Proofs

A mathematical proof is an argument that shows something is true. A proof should begin with an assumption or a given statement.

1. A statement is given.

   If two angles are supplementary to the same angle, then they are congruent to each other.

   a. What is the hypothesis?

   b. What is the conclusion?

   c. Draw a picture for the if-then statement.

The hypothesis is the given, or what is known to be true.

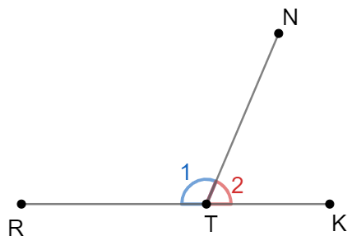
The conclusion will need to be shown, or proven, using deductive reasoning.

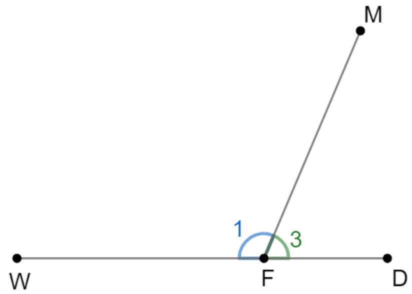
   d. Discuss with your partner why inductive reasoning is not appropriate for this context. Summarize your discussion.

One way to show the deductive reasoning that's necessary to prove the conclusion is a paragraph proof.

A **paragraph proof** is a narrative that gives the steps with justifications that prove a statement as true.

In the diagram, $\angle 1$ is supplementary to $\angle 2$, and $\angle 1$ is supplementary to $\angle 3$.





2. Work with your partner to complete the paragraph proof.

It is given that $\angle 1$ is supplementary to $\angle 2$, and $\angle 1$ is supplementary to $\angle 3$. By definition of ___________________ angles $m\angle 1 + m\angle 2 = 180°$ and ________________ $=180°$.

Since $m\angle 1 + m\angle 2 = 180°$ and $180° =$ ___________, then $m\angle 1 + m\angle 2 =$ _______________ by the transitive property. The result of applying the subtraction property of equality is $m\angle 2 =$__________. By definition of congruence, $\angle 2 \cong \angle 3$.

Another type of proof is the two-column proof.

A **two-column proof** consists of two columns of a chronological list of steps, called statements, and justifications, called reasons.

3. Complete the two-column proof.

Statement	Reason
1. $\angle 1$ is supplementary to $\angle 2$, $\angle 1$ is supplementary to $\angle 3$	1. Given
2. $m\angle 1 + m\angle 2 = 180°$ ____________ $= 180°$	2. Definition of _______________ angles
3. $m\angle 1 + m\angle 2 =$___________	3. Transitive property
4. $m\angle 2 =$_____	4. Subtraction property of equality
5. $\angle 2 \cong \angle 3$	5. Definition of congruence

If two figures have equal measurements, then by the **definition of congruence,** the figures are congruent. If congruence is shown through a series of transformations, then the **definition of congruence in terms of rigid motions** was applied.

4. Discuss with your partner why the justification for step 5 is the definition of congruence. Summarize your discussion.

5. List similarities between the paragraph proof and the two-column proof.

A **flowchart proof** shows the structure of the proof using boxes and connecting arrows. The justifications are written beside or below the box.

6. Complete the flowchart proof.

$\angle 1$ is supplementary to $\angle 2$,
$\angle 1$ is supplementary to $\angle 3$

Given

$\downarrow$

$m\angle 1 + m\angle 2 = 180°$, __________ $= 180°$

Definition of ______________ angles

$\downarrow$

$m\angle 1 + m\angle 2 =$ __________

Transitive property

$\downarrow$

$m\angle 2 =$ ________

Subtraction property of equality

$\downarrow$

$\angle 2 \cong \angle 3$

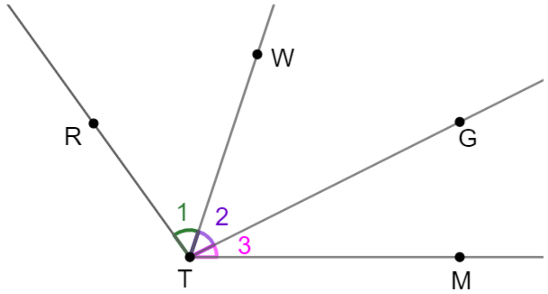
Definition of congruence

7. Which of the three types of proofs did you find the easiest to follow?

2.7.3 Guided Instruction: Flow Chart Proof

In the previous proof, each step depended on the step before it. But this is not always true.

1. Given: $m\angle 1 = m\angle 3$
   Prove:
   $\angle RTG \cong \angle WTM$



$m\angle 1 = m\angle 3$

$m\angle 1 + m\angle 2 = m\angle RTG$

$m\angle 2 + m\angle 3 = m\angle WTM$

$m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3$

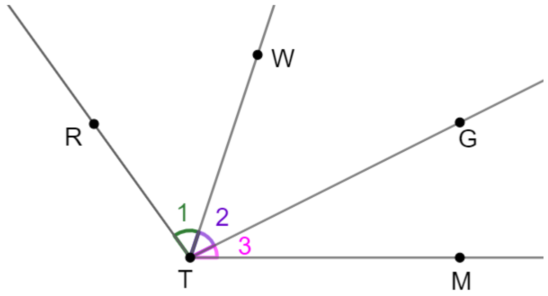
$m\angle RTG = m\angle WTM$

$\angle RTG \cong \angle WTM$

2.7.4 Guided Practice: Two-Column Proof

1. Given: $m\angle 1 = m\angle 3$
   Prove:
   $\angle RTG \cong \angle WTM$

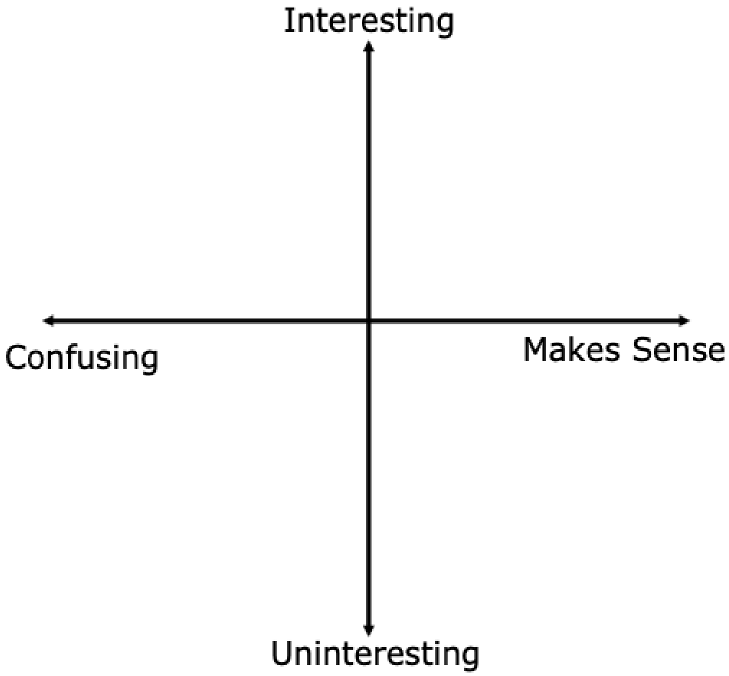
   Complete the two-column proof.



Statement	Reason
1. $m\angle 1 = m\angle 3$	1. Given
2. $m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3$	2.
3. $m\angle 1 + m\angle 2 = m\angle RTG$	3.
4. $m\angle 2 + m\angle 3 = m\angle WTM$	4.
5. $m\angle RTG = m\angle WTM$	5.
6. $\angle RTG \cong \angle WTM$	6.

2.7.5 Wrap-Up: Reflection

1. Plot a point in one of the quadrants below to indicate how you feel about today's lesson.



Unit 2, Lesson 8: Proofs – Part 2

Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.

Learning Targets

- ☐ I can reflect on the mathematical thinking found in a proof.
- ☐ I can reason about proofs using deductive reasoning, definitions, and postulates.